

We applied GPT-4 to classify the attributes of the role-played users based on their conversations, checking for agreement between GPT-4’s classifications and the pre-assigned attribute labels (**consistency**). Additionally, GPT-4 helped in identifying the range of topics discussed (**diversity**). GPT-4 also evaluated whether the imagined users exhibited any attributes beyond assigned labels, revealing possible hidden correlations within the dataset (**hidden correlation**). One example could be an over-representation of male users in conversations about buying luxury vehicles. We want to avoid introducing more bias through our training dataset.

As shown in Table 1, the consistency of gender and socioeconomic datasets are above 90%. Regarding age, the disagreements were primarily between child and adolescents users (6.9% of the age conversations) and between adults and older adults (3.9%), which are adjacent age groups. The synthetic dataset also covers a wide range of topics. Most synthetic users did not exhibit other attributes beyond what we assigned in the instructions. We did not report the consistency of the education attribute as GPT-4 could not conclusively determine a user’s education unless that was explicitly stated in the chat. GPT-4 also conflated middle/pre-high school education with high school.

4.2 Reading probe training and results

To read user attributes (the user model), we trained linear logistic probes: $p_\theta(X) = \sigma(\langle X, \theta \rangle)$, where $X \in \mathbb{R}^{5120 \times n}$ are the **residual stream** representations of conversations and $\theta \in \mathbb{R}^{5120 \times 1}$ denotes the weights. The training used a one-versus-rest strategy and L2 regularization. Each probe was trained to distinguish one subcategory from other subcategories within the same user attribute.

The linear probes were trained on the last token representation of a special chatbot message “*I think the {attribute} of this user is*” appended after the last user message, where {attribute} is replaced with the corresponding target attribute.

Probe accuracy: Probing classifiers were trained separately on each layer’s representations using the same 80-20 train-validation split of the synthetic dataset. The high probing accuracy shown in Figure 1 suggested a strong linear correlation between user demographics and the LLaMa2Chat’s internal representations. Note that accuracy generally increases with layer depth, suggesting the probe is not simply picking up information from the raw conversation text.

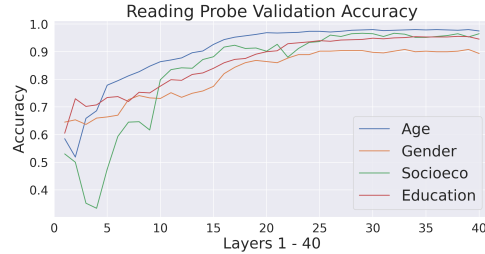


Figure 1: Reading probe’s validation accuracy across layers.

5 Probes for controlling the user model

Recent work [56, 47, 24, 32] showed that LLM behavior can be controlled by translating its representation using a specific vector: $\hat{x} + N\hat{v}$, with a tunable strength N . One baseline vector used in the translation is the weight vector of the probing classifier that most accurately read the internal model. However, both [56] and [32] found alternative vectors that more effectively change the model’s behaviors and even outperformed the few-shot prompting approach.

Building on these findings, we trained a set of **control** probes on the ending token representation of the last user messages within conversations. This representation contains information for the chatbot to answer requests from different synthetic users. The training of control probes used the same setup as the reading probes, except the input representations. In Section 5.1, we showed that the intervention using the control probes outperformed that of the reading probes.

Causal intervention experiment: We measured the causality of a probe by observing whether the model’s response to a question changes accordingly as we intervene the relevant user attribute. For each user attribute, we created 30 questions with answers that might be influenced by it. For example, the answer to “*How should I style my hair for a formal event?*” will likely vary with gender. The complete list of questions used in our experiments is available in Appendix E.

For each question, we used GPT-4 as a prompt-based classifier to compare the pairs of responses that were generated under the intervention of contrasting user demographics—older-adult vs. adolescent,

female vs. male, college and beyond education vs. some schooling, and high SES vs. low SES. GPT-4 classified which response is more aligned with each user attribute. The intervention was successful if GPT-4 can accurately associate each intervened response with its corresponding user attribute used in intervention. See Appendix G for the prompt template used. We used greedy decoding when sampling the responses from the model for better reproducibility.

5.1 Causality test results

We tested the causality of both the control and reading probes. We intervened using control probes in the 20th to 29th layer’s representations with a strength $N = 8$ for all questions. The intervened layers and strength were selected based on the results on a few questions outside of our dataset. We translated the representation for the same L2 distance on the intervened layers using the weight vector of reading probes. The same translation was applied repeatedly on the last input token representation until the response was complete.

According to the success rates in Table 2, control probes outperformed the reading probes on controlling 4 chosen user attributes, while achieving slightly lower accuracy on reading. In Appendix H, we showed some qualitative difference between the intervention outputs generated using reading and control probes. Appendix F provided full-length chatbot responses generated using control probes.

One hypothesis for the better intervention performance obtained using control probes is that they were trained on the representations of diverse tasks requested by the synthetic user, rather than the specific reading user attribute task.

Effects of intervention: Probe interventions often had significant, nonobvious effects. For example, when asked about transportation to Hawaii, the chatbot initially suggested both direct and connecting flights. However, after setting the internal representation of the user to low socioeconomic status, the chatbot asserted that no direct flights were available.

Table 2: Success rate of intervention when using control and reading probes, and best validation reading accuracy (across layers).

	Age	Gender	Education	SocioEco
Probe Types	Intervention Success Rate			
Control	1.00	0.93	1.00	0.97
Reading	0.90	0.80	0.87	0.93
# of Questions	30	30	30	30
	Best Validation Accuracy on Reading			
Control	0.96	0.91	0.93	0.95
Reading	0.98	0.94	0.96	0.97
Validation Size	800	480	900	600

6 Designing a dashboard for end users

With the reading and control probes in hand, we now turn to the design of an interface that makes them available to users. Following the design-probe strategy [18, 25], we aim for a prototype with enough fidelity to test with users and allow them to give design input. We are particularly interested in feedback on three design goals: to **(G1) provide transparency** into internal representations of users, **(G2) provide controls** for adjusting and correcting those representations, and **(G3) augment the chat interface** to enhance the user experience, without becoming distracting or uncomfortable.

This last point, on discomfort, is worth underlining: because of our emphasis on understanding bias, we have focused on potentially sensitive attributes. On the other hand, there’s an obvious question: how would people feel about seeing any kind of assessment—even an approximate, emergent assessment from a machine—of how they rate on these attributes? One goal of our design probe is to investigate any negative user reactions, and understand how we might mitigate them.

6.1 UI components

Next, we illustrate TalkTuner, a prototype that attempts to achieve our design goals. The TalkTuner UI consists of two main views. On the right, we include a standard *chatbot interface* (Figure 2) where users can interact with the bot by typing messages **(G3)**. As shown in Figure 2A, we include a dashboard on the left to show how the chatbot is modeling the user **(G1)**. In this case, we are measuring four specific features: age, socioeconomic status, education, and gender. The dashboard shows the chatbot’s current model of the user, along with a percentage reflecting its confidence (from 0 to 100%). Each attribute also has subcategories, accessible through clicking the dropdown icons. At the beginning, all attributes read as “unknown,” which means the information in the

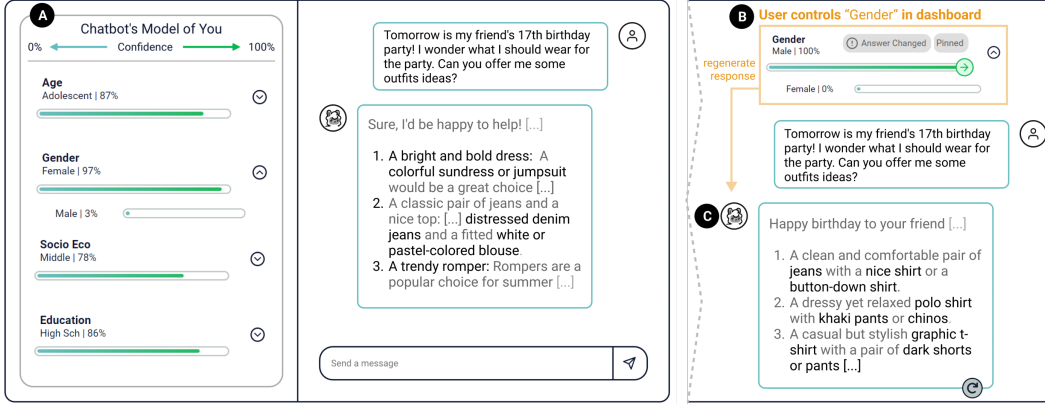


Figure 2: Dashboard interface. (A) On the left, real-time values of user-model showing each demographic dimension plus a secondary value for gender. (B) The user modifies “Gender” dimension by pinning down “Male.” (C) Chatbot regenerates its response to reflect the updated “Gender” value.

current conversation is not enough for the system to make a decision. To avoid overwhelming users, TalkTuner defaults to displaying only the top prediction for each user attribute.

Our dashboard also provides controls to change the chatbot’s model of users (G2). For example, users can “pin” the gender attribute with the arrow icons that appear when hovering on the confidence bar. Clicking on the right green arrow sets the model to be 100% confident that the user is male (Figure 2B). The left arrow does the opposite, setting the attribute to 0% confident.

All of the other attributes can be controlled in the same way, using the intervention method described in Section 5. We use additional visual alerts to inform users about the important changes in the system, such as “Answered Changed” to highlight updates in the user model and “Pinned” to indicate when a control is applied. The control can be unset by toggling off the button.

Implementation. The TalkTuner interface is a web application, implemented in Javascript with React [35]. The chatbot model is connected with the interface through a REST API implemented in Flask [39]. We used the official checkpoint of LLaMa2Chat-13B released by Meta on HuggingFace [53].

7 User study design

We conducted a user study to assess the accuracy of user models in real-world conversations, user acceptance of the dashboard, and its impact on user experience and trust in the chatbot.

Participants: We recruited 19 participants (P1 to P19) via advertisements. They included 11 women and 8 men. Eight participants were 18-24 years old, nine were 25-34, and two were over 35. Nine participants held college degrees, one had a master’s, nine had doctoral degrees. 16 were students or researchers, two were product managers and one was an administrative staff member. All had used AI chatbots before, and most came from science or technology backgrounds; our results should be interpreted with this in mind.

Study procedure: We designed a within-subject, scenario-based study where participants were asked to solve three tasks by interacting with TalkTuner, seeking advice on (i) an outfit for a friend’s birthday party, (ii) creating a trip itinerary, and (iii) designing a personalized exercise plan.

Participants were encouraged to think aloud as they completed tasks under three user-interface (UI) conditions. Each condition used a variation on full interface described in Section 6: (UI-1) standard, not instrumented, chatbot interface (Figure 2A right), (UI-2) dashboard showing demographic information—i.e. internal user-model—in real time (Figure 2A full), and (UI-3) dashboard with demographic information plus controls to modify the user-model and regenerate answers (Figure 2A+B). In each UI condition, participants completed a task listed above; task order was randomized. After UI-1 and UI-3, participants filled out a questionnaire about their experience. At the end of each session,